Week 4 Homework

This is a preview of the published version of the quiz

Started: Jul 2 at 7:48am

Quiz Instructions

Question 1

1 pts

(Lesson 3.1: Solving a Differential Equation.) Suppose that $f(x) = e^{2x}$. We know that if h is small, then
$f'(x) \approx \frac{f(x+h)-f(x)}{h}$.
Using this expression with $h = 0.01$, find an approximate value for $f'(1)$.

- a. 1
- b. 2.72
- c. 7.38
- d. 14.93

Question 2

1 pts

(Lesson 3.1: Solving a Differential Equation.) Suppose that $f(x) = e^{2x}$. What is the actual value of $f'(1)$?

- a. 1
- b. $e \approx 2.72$
- c. $e^2 \approx 7.39$
- d. $2e^2 \approx 14.78$
- e. 14.93

Question 3

1 pts

(Lesson 3.1: Solving a Differential Equation.) Consider the differential equation $f'(x) = (x+1)f(x)$ with $f(0) = 1$. What is the exact formula for $f(x)$?

- a. $f(x) = e^x$
- b. $f(x) = e^{2x}$
- c. $f(x) = \exp\left\{\frac{x^2}{2} + x\right\}$
- d. $f(x) = \exp\{x^2 + 2x\}$

Question 4

1 pts

(Lesson 3.1: Solving Differential Equations.) Consider the differential equation $f'(x) = (x+1)f(x)$ with $f(0) = 1$. Solve for $f(0.20)$ using Euler's approximation method with increment $h = 0.01$ for $x \in [0, 0.20]$.

- a. $f(0.20) \approx 0.0$
- b. $f(0.20) \approx 1.0$
- c. $f(0.20) \approx 1.24$
- d. $f(0.20) \approx 2.49$

Question 5

1 pts

(Lesson 3.2: Monte Carlo Integration.) Suppose that we want to use Monte Carlo integration to approximate $I = \int_1^3 \frac{1}{1+x}\, dx$. If $U_1, U_2, \ldots, U_n$ are i.i.d. Unif(0,1)'s, what's a good approximation $\bar{I}_n$ for I?

- a. $\frac{1}{n}\sum_{i=1}^{n}\frac{1}{1+U_i}$
- b. $\frac{2}{n}\sum_{i=1}^{n}\frac{1}{1+U_i}$
- c. $\frac{1}{n}\sum_{i=1}^{n}\frac{1}{1+2U_i}$
- d. $\frac{2}{n}\sum_{i=1}^{n}\frac{1}{1+2U_i}$
- e. $\frac{1}{n}\sum_{i=1}^{n}\frac{1}{1+3U_i}$

Question 6

1 pts

(Lesson 3.2: Monte Carlo Integration.) Again suppose that we want to use Monte Carlo integration to approximate $I = \int_1^3 \frac{1}{1+x}\,dx$. You may have recently discovered that the MC estimator is of the form $\bar{I}_n = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{1+U_i}$.

Estimate the integral I by calculating $\bar{I}_n$ with the following 4 uniforms:

$$U_1 = 0.3 \qquad U_2 = 0.9 \qquad U_3 = 0.2 \qquad U_4 = 0.7$$

- a. 0
- b. 0.2
- c. 0.321
- d. 0.679
- e. 0.8

Question 7

1 pts

(Lesson 3.2: Monte Carlo Integration.) Yet again suppose that we want to use Monte Carlo integration to approximate $I = \int_1^3 \frac{1}{1+x}\,dx$. What is the *exact* value of I?

- a. 0.197

- b. 0.693
- c. 1.386
- d. 2.773

Question 8

1 pts

(Lesson 3.3: Making Some π.) Inscribe a circle in a unit square and toss $n = 1000$ random darts at the square. Suppose that 760 of those darts land in the circle. Using the technology developed in class, what is the resulting estimate for π?

- a. π
- b. 4.0 (UGA answer)
- c. 3.2
- d. 3.04
- e. 3.12

Question 9

1 pts

(Lesson 3.4: Single-Server Queue.) Consider a single-server Q with *LIFO* (*last*-in-first-out) services. Suppose that three customers show up at times 5, 6, and 8, and that they all have service times of 4. When does customer 2 leave the system?

- a. 3
- b. 9
- c. 13
- d. 17
- e. 19

Question 10

1 pts

(Lesson 3.5: (s, S) Inventory Model.) Consider our numerical example from the lesson. What would the third day's total profits have been if we had used a (4,10) policy instead of a (3,10)?

- a. -22
- b. -13
- c. 44
- d. 45
- e. 70

Question 11

1 pts

(Lesson 3.6: Simulating Random Variables.) If U is a Unif(0,1) random number, what is the distribution of $-0.5\ell n(U)$?

- a. Who knows?
- b. Exp(2)
- c. Exp(1/2)
- d. Exp(-2)
- e. Exp$(-1/2)$

Question 12

1 pts

(Lesson 3.6: Simulating Random Variables.) If U_1 and U_2 are i.i.d. Unif(0,1) random variables, what is the distribution of $U_1 + U_2$? Hints: (i) I may have mentioned this in class at some point; (ii) You may be able to reason this out by looking at the

distribution of the sum of two dice tosses; or (iii) You can use something like Excel to simulate $U_1 + U_2$ many times and make a histogram of the results.

- a. Unif(0,2)
- b. Normal
- c. Exponential
- d. Triangular

Question 13

1 pts

(Lesson 3.7: Spreadsheet Simulation.) I stole this problem from the Banks, Carson, Nelson and Nicol text (5th edition). Expenses for Joey's college attendance next year are as follows (in $):

Tuition = 8400
Dormitory = 5400
Meals $\sim$ Unif(900,1350)
Entertainment $\sim$ Unif(600,1200)
Transportation $\sim$ Unif(200,600)
Books $\sim$ Unif(400,800)

Here are the income streams the student has for next year:
Scholarship = 3000
Parents = 4000
Waiting Tables $\sim$ Unif(3000,5000)
Library Job $\sim$ Unif(2000,3000)

Use Monte Carlo simulation to estimate the expected value of the loan that will be needed to enable Joey to go to college next year.

- a. $2500
- b. $3250
- c. $3325
- d. $3450

e. $4000

Not saved

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